



High-thrust ultralight ion wind engine toward controllable aerial microrobots

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ABSTRACT

The ion wind engine, an innovative propulsion system, transforms ion energy into mechanical thrust, offering unique advantages such as low noise, no moving parts, a compact form factor, and a superior thrust-to-weight ratio. This technology shows exceptional promise for powering miniaturized robotic systems, particularly one of the most challenging types—flying microrobots. In this study, we developed a three-dimensional electrohydrodynamic (EHD) model to investigate a needle-to-plane ion wind engine, integrating the coupled effects of electrostatic fields, ion transport, and fluid dynamics. Using simulation-driven insights, we optimized the engine's structure, achieving enhanced performance with a four-collector configuration weighing 18.1 mg. Experimental validation through Schlieren imaging and thrust measurements confirmed high wind speeds and a thrust of 0.66 mN, yielding a thrust per unit volume of 261 N/m³ and a thrust-to-weight ratio of 3.7. As a proof-of-concept, we designed and fabricated a microrobot powered by the optimized ion wind engine, demonstrating precise flight attitude control via independent engine operation. These findings highlight the potential of multi-engine ion wind systems for advancing microrobot flight dynamics, with significant implications for the design of next-generation microscale aerial vehicles.

1. Introduction

Over the past decade, ion wind engines, which convert ion energy into mechanical thrust, have attracted increasing attention as an innovative propulsion technology [1–3]. These systems operate by generating airflow through corona discharge between electrodes under high voltage, where ions are accelerated in an electric field to produce thrust [4]. Due to their inherently simple design, free of complex mechanical components, ion wind engines offer stable and reliable operation at the microscale. Moreover, the ion wind engine is characterized by a high thrust-to-weight ratio, low power consumption, and silent operation [5,6]. These advantages make ion wind engines particularly well-suited for applications in miniaturized robotics. As a result, the study of microrobots powered by ion wind has emerged as a promising and rapidly evolving area of research.

Compared to the rotary-wing propulsion system [7,8] and the flapping-wing propulsion system [9,10], the ion wind engine offers distinct advantages. Unlike the rotary-wing propulsion system, they have no motors or moving parts, and unlike the flapping-wing propulsion system, they do not rely on high-frequency flapping mechanisms.

Rotary-wing and flapping-wing systems primarily generate lift through aerodynamic effects, whereas ion wind engines operate based on a fundamentally different principle. In these systems, a high-voltage field is applied to the emitter electrode with a small radius of curvature, creating a strong electric field at its sharp tip that ionizes surrounding air molecules. The ions produced are rapidly accelerated by the electric field, triggering further ionization through collisions with neutral molecules. This cascade of ionization generates a dense stream of ions, which are propelled toward the collector electrode. The momentum exchange between the accelerated ions and neutral air molecules produces ion wind. Since the discovery of the EHD phenomenon, scientists and engineers have conducted in-depth research, gradually elucidating the mechanisms behind ion wind generation [5]. Recently, ion wind has been widely applied across various fields, including pumps [11,12], sensors [13,14], electrostatic precipitators [15,16], aerodynamic flow control [17,18], and electric propulsion [1,19–21]. Ion wind propulsion remains promising even in low-density environments, such as the stratosphere or on Mars. With careful optimization of device structure

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and electrode spacing, it is theoretically possible to achieve flight in such challenging conditions [22,23].

Owing to the advantages of ion wind systems, researchers have conducted studies and optimization designs based on simulation and experimental methods. For instance, Moreau et al. [24] developed a platform combining electrical measurements, Intensified Charge Coupled Device (ICCD) imaging, and Particle Image Velocity (PIV) measurements to characterize ion wind generation from corona discharges. Similarly, Zhang et al. [12] integrated multiphysics simulations with experimental validation to perform multi-parameter optimization of ion wind pumps. Their work addressed non-uniform flow velocity distribution in early designs and significantly improved both outlet velocity and flow uniformity. Coseru et al. [25] employed an electroaerodynamic (EAD) model using a Newton–Raphson iterative method with adaptive meshing to achieve convergence. Their approach enabled the prediction of thrust and current, leading to the identification of optimal electrode configurations. Researchers have conducted studies on various configurations of ion wind systems using a combination of experimental and computational methods. Common configurations include needle-to-plate [26], needle-to-ring [6], wire-to-plate [19], and wire-to-cylinder [27]. The studies demonstrated that different factors, such as structure, spacing, material, temperature, and humidity can influence wind speed and thrust [12,28,29]. The discharge characteristics also contribute to performance of ion wind. A previous study revealed that negative corona discharges produce more stable ion flows, while positive discharges generate faster but more turbulent flows, especially when transitioning to a spark regime [24]. Several specialized designs with distinct advantages have been developed based on these basic configurations. Johnson et al. [30] designed two flat plates operating as a gate electrode and a downstream electrode to reduce the corona onset voltage. Dau et al. [31,32] designed parallel pin geometry to generate neutralized ion wind from two emitter electrodes of opposite polarity. Hou et al. [33] explored innovative electrode geometries, such as sawtooth emitters and multi-ring collectors, to enhance thrust performance, demonstrating the effectiveness of geometric optimization in improving ion wind output. Moreover, some subsequent studies have centered on the merits of multi-stage ion wind propulsion systems. It has been demonstrated that the integration of multiple stages or cascaded configurations can significantly enhance thrust performance by optimizing the energy transfer between successive stages [34,35].

Significant milestones of ion-propelled aircraft have been achieved in recent years, demonstrating the feasibility of the ion wind engine for flight. The team from the Massachusetts Institute of Technology successfully conducted the first sustained flight of an ion-propelled fixed-wing aircraft [1]. The aircraft had a wingspan of 5 m and a total mass of 2.45 kg. Powered by a custom battery and a high-voltage power converter capable of stepping up the battery voltage from 200 V to 40 kV, the system generated 3.2 N of thrust, enabling a sustained 60-m indoor flight. In the field of micro-robotics, the team from the University of California, Berkeley developed a 30 mg ion-propelled micro drone capable of controlled flight while carrying a 37 mg inertial measurement unit (IMU). The system operated using an externally supplied high-voltage source, successfully showcasing a flight of a robot in centimeter scale [20]. Additionally, various methods for controlling ion wind thrust have been explored. These control strategies include adjusting the voltage difference across electrodes to modulate the thrust [20], manipulating the relay switching frequency to control the thrust output [36], and even leveraging magnetic fields to influence the ion wind within the system [37]. Each of these methods offers a unique approach to fine-tuning the ion wind propulsion for a specific purpose. However, the researchers have only demonstrated the feasibility of controlled flight through simulations [20] or component tests [36], without achieving actual control of the robot's attitude. At present, centimeter-scale microrobots capable of controlling flight attitude primarily rely on flapping wings for propulsion [9,38]. Given this,

it is crucial to explore ion wind engines as an alternative, solid-state propulsion method for controlled flight in microrobots.

Despite prior studies on the mechanisms, structural optimization, and control strategies of ion wind engines, a critical gap remains in the analysis and optimization of microrobots equipped with multiple independently controllable ion wind engines as an integrated system. This research is crucial for advancing ion wind propulsion for real-world applications across diverse systems, particularly microscale aerial vehicles. This study presents the design, simulation, optimization, fabrication, and experimental validation of multiple ion wind engines for microrobotics applications, demonstrated through a flying microrobot powered by EHD propulsion. A needle-to-plane electrode configuration was adopted for the construction of a miniaturized ion wind engine due to the simplicity of its manufacturing process and the flexibility of its design. The engine's structural parameters are optimized using multiphysics finite element analysis, incorporating coupled electrostatic fields, ion transport, and fluid dynamics, providing insights into the optimal design of the ion wind propulsion system. The engine is designed to achieve a higher thrust-to-weight ratio under lower voltage, meeting stringent performance requirements and ensuring efficient operation for microrobotics applications. The effectiveness of the optimized structure is validated experimentally through thrust measurement tests and Schlieren photography tests. Furthermore, two different configurations of the microrobot are investigated through three-dimensional simulations, leading to the preliminary development of a more efficient microrobot configuration. The feasibility of independently controlling each ion wind engine is thoroughly demonstrated, offering a significant improvement in maneuverability and realizability of attitude control. Additionally, the microrobot prototype capable of liftoff has been successfully fabricated. Experimental testing of the prototype confirms that the design works as intended and that the control approach effectively manages the robot's attitude. These tests demonstrate the practicality of the design and highlight its potential for real-world applications, where precise control and flexibility are essential.

2. Design of the ion wind engine

2.1. Theoretical analysis

An ion wind engine consists of emitters and collectors. The voltage difference applied between the emitter and the collector generates ion wind, based on the mechanism of EHD. EHD primarily describes the movement of diffused charges under the influence of an externally applied electric field. A theoretical model for one-dimensional ion wind propulsion expresses the thrust F_t as [39]:

$$F_t = \int \rho_e E d\Omega = \frac{Id_g}{\mu} \quad (1)$$

where ρ_e is the space charge density, E is the electric field strength, Ω is the volume, I is the ion current, d_g is the electrode gap distance, and μ is the ion mobility in air. From Eq. (1), it is evident that an increase in current I and distance d_g both contribute to an enhancement in thrust. The current can be expressed with the Townsend relation, which describes the voltage–current characteristics of corona discharge [39,40]:

$$I = \frac{C'\mu V(V - V_i)}{d_g^2} \quad (2)$$

where V_i is the corona onset voltage, and C' is an empirical constant determined by the geometry of the electrodes. By combining Eq. (1) and (2), the EHD thrust can be further derived as:

$$F_t = \frac{C'\mu V(V - V_i)}{d_g} \quad (3)$$

Based on Eq. (3), to maximize this force, the operating voltage V should approach the breakdown voltage V_B . According to Paschen's Law [41], the breakdown voltage is given by:

$$V_B = \frac{Bpd_g}{\ln(Apd_g) - \ln\left[\ln\left(1 + \frac{1}{\gamma}\right)\right]} \quad (4)$$

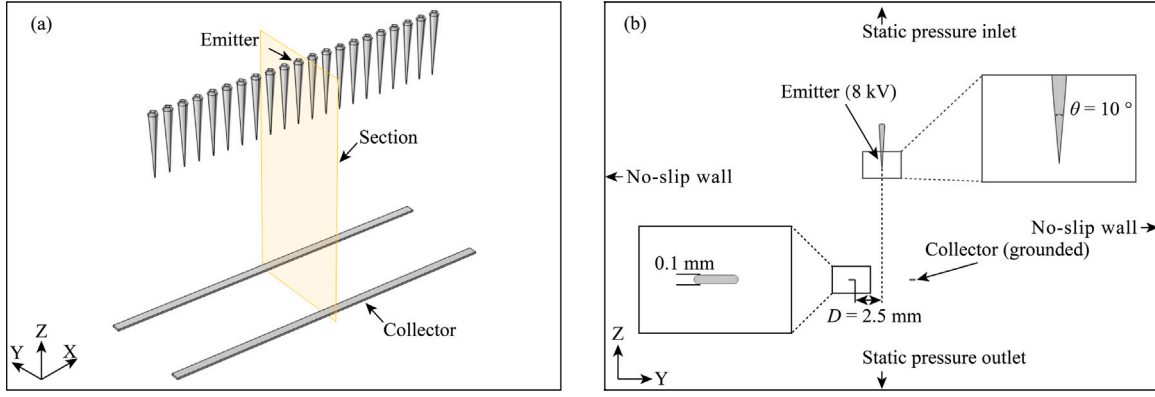


Fig. 1. Illustration of the physical model. (a) Three-dimensional model of the ion wind engine. (b) Cross-sectional figure of the physical model and the detailed boundary conditions.

where p is the gas pressure, γ is the secondary electron emission coefficient of the cathode, and A and B are empirical constants. These constants are related to temperature and the properties of the medium, and the electrodes. In the air medium, when the distance d_g is greater than 0.1 mm, the breakdown voltage increases as the gap widens [42, 43]. However, the increase in the electrode gap distance leads to an increase in the corona onset voltage. Furthermore, an excessive increase in d_g has the potential to adversely affect flight stability due to a raised center of gravity. In this study, the electrode spacing is set to 10 mm to achieve a balance between lift enhancement and flight stability.

For an ion wind engine used in a micro aerial vehicle, the thrust-to-area ratio can be derived from Eq. (1) as:

$$\frac{F_t}{S} = \frac{j d_g}{\mu} \quad (5)$$

where S is the area of the electrode, and j is the current density. The thrust-to-power ratio is also derived as:

$$\frac{F_t}{P} = \frac{F_t}{IV} = \frac{\Delta p}{jV} \quad (6)$$

where P is the input electrical power, Δp is the pressure difference induced by EHD. By applying the Bernoulli equation, the thrust can also be expressed more simply as [44]:

$$F_t = \frac{1}{2} S \rho_f v^2 \quad (7)$$

where ρ_f is the density of the neutral gas and v is the flow velocity of the neutral fluid. From Eq. (7), thrust is positively correlated with the flow velocity of the neutral fluid. Thus, the optimal design of an ion wind engine with higher thrust can be preliminarily determined by evaluating the calculated flow velocity of different structures.

2.2. Simulation modeling and results

Even though the basic physical characteristics of ion wind can be obtained through the above one-dimensional analysis, more detailed flow dynamics is required for the design of the engine. This study employs the simulation method based on fully coupled multiphysics finite element analysis (COMSOL Multiphysics 6.2), to analyze the ion wind engine with both two-dimensional and three-dimensional models. Initially, an ion wind engine based on the needle-surface structure was designed. A row of needles with a height of 4 mm serves as the emitter, while a rectangular plate with a width of 0.5 mm and a thickness of 0.1 mm acts as the collector. The vertical distance from the needle tips to the collector is 10 mm. In the preliminary design, the needle tip angle is set at 10° , and two collector electrodes are positioned at a distance of 5 mm horizontally from each other. The three-dimensional physical model is illustrated in Fig. 1(a).

The flow of the ion wind is fundamentally modeled by the interaction of distinct physical fields, involving the electric field, the charge field, and the flow field. The specific method involves using the electrostatic module, transport of diluted species module, and laminar flow module in COMSOL software to solve the Poisson's equation, charge transport equation, and the Navier–Stokes equations, respectively.

A corona discharge is generated when a high voltage is applied to the emitter. The charge conservation equations, which describe the electric field and charge density, are derived from Poisson's equation as follows [45]:

$$\mathbf{E} = -\nabla V \quad (8)$$

$$\nabla \cdot (\epsilon \mathbf{E}) = \rho_e \quad (9)$$

where $\mathbf{E}$ is the electric field strength, ρ_e is the space charge density and ϵ is the permittivity of the medium. Then the ions formed by the ionization of air move toward the collector electrode, generating a current. The transport of charged particles in space follows the Nernst–Planck equation, which describes the diffusion, migration, and convection of the particles. The governing equations used here are:

$$\nabla \cdot \mathbf{J} + \mathbf{U} \cdot \nabla c = R \quad (10)$$

$$\mathbf{J} = -D \nabla c - z u_m F_i c \nabla V \quad (11)$$

where $\mathbf{J}$ is the current density, c is the ion concentration, $\mathbf{U}$ is the velocity field of the ion wind, R denotes the rate of change of ion concentration, D is the ionic diffusion coefficient, z is the charge number of the ion, F_i is the Faraday constant, and u_m is the ionic mobility, which can be calculated by Nernst–Einstein relation.

The airflow generated under a high-voltage electric field can be regarded as an incompressible fluid. The motion of the air is governed by the Navier–Stokes equation. For the steady-flow phase, the Navier–Stokes equation is simplified to:

$$\rho(\mathbf{U} \cdot \nabla) \mathbf{U} = -\nabla p + \eta \nabla^2 \mathbf{U} + \mathbf{F}_e \quad (12)$$

where ρ is the density of the air, η is the kinematic viscosity, and $\mathbf{F}_e$ is the EHD force acting on the air.

A simplified schematic of the structure in the XOZ projection plane, along with the detailed boundary conditions, is shown in Fig. 1(b). The boundary conditions related to the electrostatic field are as follows: an electric potential of 8 kV is applied to the emitter; the collectors are grounded; and the rest are set to zero charge. The boundary conditions related to the species transport are as follows: part of the needle tip is set to allow the flow of the diluted species with a certain concentration into the computational domain; while the collectors are set to the flow-out boundary. Regarding the fluid flow boundary conditions, the upper side of the computational domain is set to the static pressure inlet, the

Table 1
Physical parameters in simulation.

Parameters	Unit	Value
Diffusion coefficient	$\text{m}^2 \text{s}^{-1}$	5.3×10^{-5}
Air density	kg m^{-3}	1.2
Relative dielectric constant of air		1
Dynamic viscosity of air	Pa s	1.8×10^{-5}

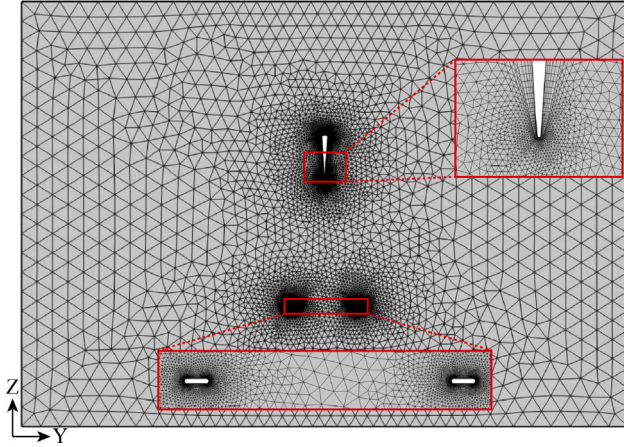


Fig. 2. The mesh diagram in COMSOL Multiphysics.

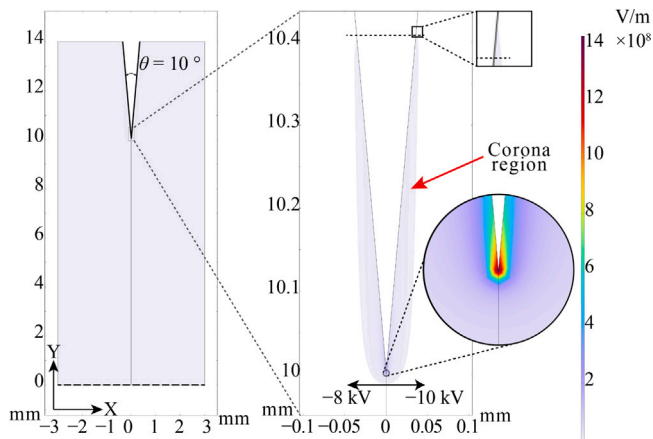


Fig. 3. Simulated electric field intensity near the emitter tip under different voltage differences.

lower side of the computational domain is set to the static pressure outlet, and the boundaries on both sides are set to the no-slip wall. The simulation parameters are listed in Table 1.

Then the computational domain of the model shown in Fig. 1(b) is meshed, as illustrated in Fig. 2. To improve modeling efficiency, an unstructured mesh is used to discretize the computational domain. However, to more accurately capture the variations in fluid velocity and electric field strength near the emitters and collectors, a structured mesh with finer resolution is employed in these regions. The mesh near these areas is further refined, as illustrated. Additionally, the number of boundary layers near the needle tip is set to five.

The corona zone near the emitter tip is critical and is analyzed to determine the range of boundary for the inflow of the diluted species in the calculation of the transport of diluted species. The corona zone is determined through the application of Poisson equations, utilizing the local field approximation [46,47]. The distribution of the electric field intensity near the needle tip under 8 kV is shown in Fig. 3.

According to the simulation results, only the electric field strength in regions where the value exceeds $1.5 \times 10^7 \text{ V/m}$ (assumed to be the

corona discharge initiation threshold) is displayed in Fig. 3. The high potential difference that is applied between the emitters and collectors leads to the accumulation of ions at the sharply curved tip due to the small radius of curvature. This results in a local electric field intensity that exceeds the ionization threshold of the surrounding gas. The ions are generated at the tip of the needle, and the electric field created by the potential difference accelerates the ions. Then the ions collide with neutral air molecules to generate ion wind. It can be seen from Fig. 3 that the electric field is highly concentrated at the sharply curved tip, where the field strength reaches its maximum. In contrast, the surrounding regions exhibit significantly lower intensities, indicating a rapid spatial decay of the electric field away from the tip. As the voltage difference increases, the coverage of the corona zone expands, while the overall distribution pattern remains unchanged. Upon attaining an applied voltage of 8 kV, the height of the corona zone extends up to 10.35 mm. According to the Kaptzov hypothesis [47], once the applied voltage exceeds the corona onset voltage, the electric field strength at the surface of the corona electrode remains constant. This assumption simplifies the analysis by neglecting complex particle dynamics within the corona layer and assuming that ions are emitted directly from the surface of the electrode. Based on this assumption, only a localized surface near the needle tip is selected to assign the ion concentration.

The subsequent task is to ascertain the mesh diagram in COMSOL Multiphysics for the simulation of the velocity of the ion wind generated by the ion wind engine. Fig. 2 shows the refined mesh diagram in COMSOL Multiphysics. The wind speed distribution obtained from simulations with different mesh densities exhibits similar curves, as illustrated in Fig. 4. The primary ion wind region is observed to extend downward from the emitter needle tip, with the concentration located between the two collectors. The primary direction of the ion wind is along the negative Z-axis. After the ion wind passes over the collectors, the wind speed decreases, expanding toward the sides. The position of maximum wind speed is located below the collector plane. To evaluate the impact of the mesh on the simulation results, four densities were compared: coarse, normal, refined, and extremely fine. The number of mesh elements for each case was 5,070, 8,560, 16,892, and 59,670, respectively. The velocity of the ion wind was extracted at a depth of 5 mm beneath the collector to ascertain the independence of the mesh. Although the number of elements in the refined mesh and the extremely fine mesh increased by 2 times and 6 times, respectively, compared to the normal mesh, the calculated velocity is only 1.56% and 1.89% higher, as presented in Fig. 5. Therefore, to balance computational accuracy and efficiency, the refined mesh is adopted for subsequent simulations in three dimensions. The boundary conditions in the three-dimensional simulations are consistent with those used in the two-dimensional simulations.

2.3. Parameter optimization and structural design

Following the establishment of the simulation model, the effects of key parameters on ion wind engine performance were analyzed to optimize its design. The velocity of the ion wind generated by the engine was selected to characterize the thrust performance of the engine based on Eq. (7). Using the model shown in Fig. 1 as the base model and applying a voltage difference of 8 kV, the relationship between the velocity of ion wind and the angle of the needle tip was first investigated. As demonstrated in Fig. 6(a), the decrease in tip angle results in a higher charge density at the needle tip, consequently generating a higher electric field intensity near the tip. The higher electric field intensity, in turn, facilitates the ionization of surrounding air molecules, increasing the ion wind velocity. The velocity was also calculated for varying needle tip angles with electrode-to-centerline distances D of 2.5 mm and 5 mm. It can be observed that as the distance between two collectors D increases, the velocity decreases. Smaller tip angles increase the velocity of the ion wind and facilitate lightweight

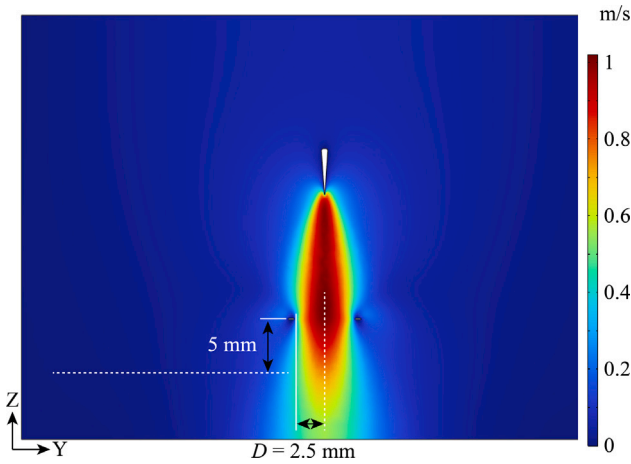


Fig. 4. The simulated velocity of ion wind under 8 kV.

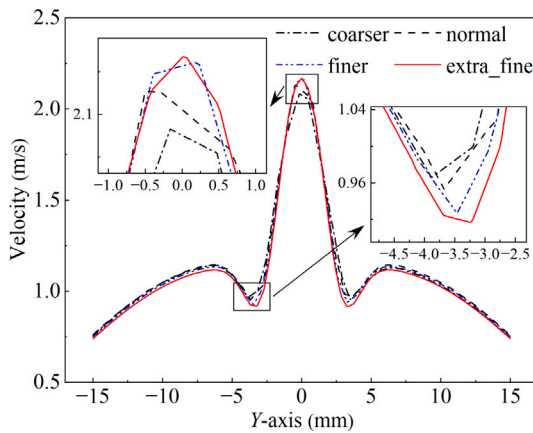


Fig. 5. The simulated velocity of ion wind at a depth of 5 mm beneath the collector based on meshes with different levels of refinement.

design. However, manufacturing limits and reduced corrosion resistance of needles with a sharper tip lead to the selection of a 10° uniform tip angle.

The number of needle tips also affects the velocity of the ion wind generated by the ion wind engine. The number of needle tips in the emitter within a fixed length (18 mm) was varied to analyze the influence. The simulation results shown in Fig. 6(b) indicate that as the number of tips increases, both the maximum and average velocity increase linearly with the number of tips. However, considering the discharge range calculated in Fig. 3, excessive tip density can disturb the local electric field due to mutual interference between neighboring tips, leading to partial field shielding, which in turn reduces discharge efficiency and wind velocity. Taking into account the discharge range at the tip, a design with 21 densely packed tips arranged at a 10° angle and a cathode length of 18 mm was selected.

In addition to the emitter configuration, the spatial arrangement of the collectors also significantly affects wind speed and thrust due to changes in the paths of the ion migration. The arrangement where the collector is directly aligned with the emitter yields the shortest migration path and minimizes kinetic energy loss [19,34,48]. However, because the wind generated by EHD is directly in a collision with the collector, a significant amount of energy is dissipated. In contrast, the symmetric arrangement of more collectors leaves space directly beneath the emitter, as shown in Fig. 1. This configuration facilitates the establishment of a flow channel for the airflow, thereby generating higher thrust.

Table 2

Geometric parameters of the optimized ion wind engine.

Parameter	Unit	Value
Emitter size	mm $\times$ mm $\times$ mm	$0.1 \times 4 \times 20$
Number of needle tips in emitter		21
Tip angle of emitter	$^\circ$	10
Distance from tip of emitter to collector mm	10	
Collector size	mm $\times$ mm $\times$ mm	$0.1 \times 0.5 \times 20$
Outer collector distance to axis	mm	5.5
Inner collector distance to axis	mm	2.5

Furthermore, an attempt is made to further increase the number of collector electrodes to achieve higher thrust. To determine the optimal path in a symmetric configuration, two pairs of $20 \text{ mm} \times 0.5 \text{ mm} \times 0.1 \text{ mm}$ plates were symmetrically arranged as collectors, as illustrated in Fig. 7(a). The distance between one of the outer collectors and the central axis d_{out} was varied from 5.5 mm to 2 mm in 0.5 mm steps, while the distance between one of the inner plates and the central axis d_{in} was scanned from 4.5 mm to 1.5 mm, also in 0.5 mm steps.

The simulation results of the maximum velocity of the ion wind on the plane 5 mm below the collector are plotted in Fig. 7(b). The maximum velocity of the configuration that contains only one pair of collectors is also plotted in Fig. 7(b) for comparison. Structures with the inner collectors placed more than 2 mm away from the central axis generally outperform the configuration with single-pair collectors. Overall, for configurations in which the inner collectors are placed more than 2 mm from the central axis, increasing the distance between the outer collectors and the central axis leads to a slight improvement in wind speed. However, when the inner collectors are placed farther away – approaching half the distance between the needle tip of the emitter and the plane of collectors – the ion wind speed tends to decrease instead. According to the simulation results, when the inner and outer collectors are placed at distances of 2.5 mm and 5.5 mm from the central axis, respectively, the configuration yields a maximum ion wind velocity of 1.255 m/s. We also estimated the thrust by integrating the wind velocity at the cross-section near the lower surface of the collector. The thrust for the engine with four collectors was calculated to be 0.17 mN, which is higher than the 0.13 mN calculated for the engine with two collectors.

Through the above analysis of the ion wind engine, the geometric parameters of the optimized structure are determined, as listed in Table 2. The simulation results of the electric potential field and the velocity field are shown in Fig. 8. When the applied voltage is 8 kV, in the ellipsoid-shaped region near the emitter, the electric potential exceeds 4 kV. For clarity, the ion wind velocity distribution is shown only in the regions where the wind speed exceeded 0.5 m/s, with high-speed regions primarily concentrated beneath the emitter and a maximum velocity of 1.52 m/s occurring at the structure's geometric center, as presented in Fig. 8(b).

3. Experimental validation of the ion wind engine

Following the analysis of each individual configuration parameter, a set of design factors was identified to develop an improved ion wind engine. Subsequently, experimental investigations of the velocity and thrust characteristics were conducted to validate the performance enhancements achieved through this optimization. The emitter and the collector of the ion wind engine were made using 100- μm -thick polyethylene terephthalate (PET) film with aluminum coating. The film was precisely cut by the ultraviolet (UV) laser cutter with a cutting accuracy of 10 μm . The connectors to connect the emitter and collectors were cut from the nonconducting PET film. The structure, assembled using UV-curable epoxy resin, was investigated for the ion wind airflow in two-collector and four-collector ion wind engine designs using the Schlieren photography technique. The test system and the fabricated

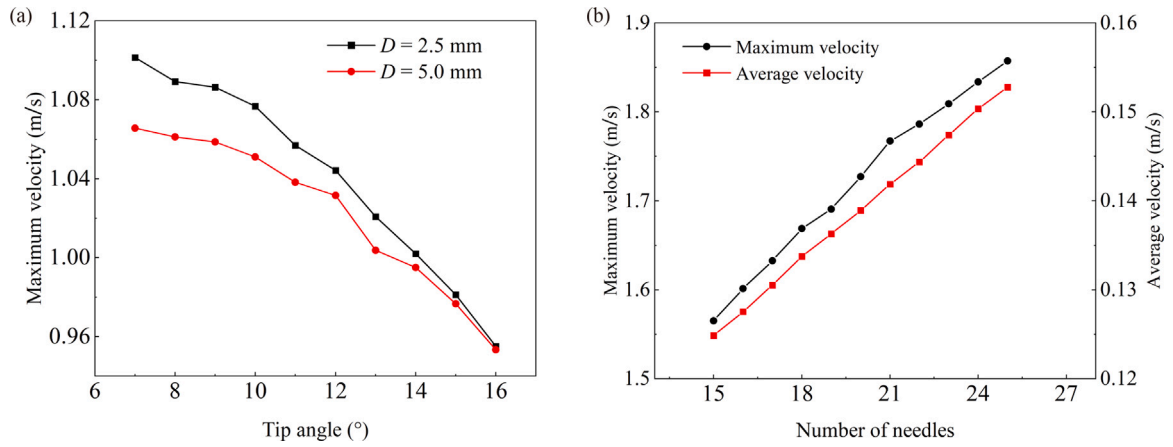


Fig. 6. The maximum velocity of ion wind versus (a) the angle of the needle tip, and (b) the number of the needles.

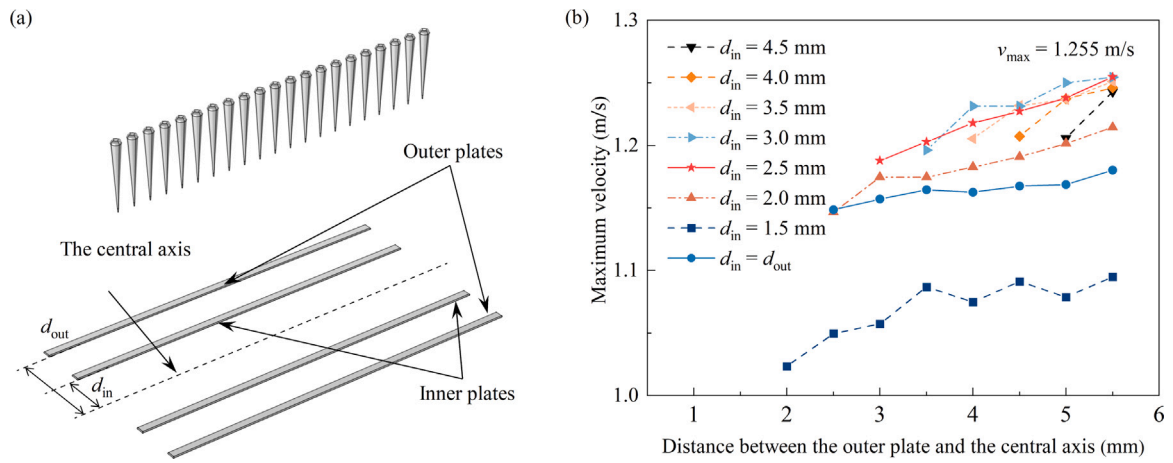


Fig. 7. Distribution of the ion wind velocity under different collector configurations. (a) Schematic diagram of the structure of the plate arrangement. (b) Maximum velocity of the ion wind generated by different configurations.

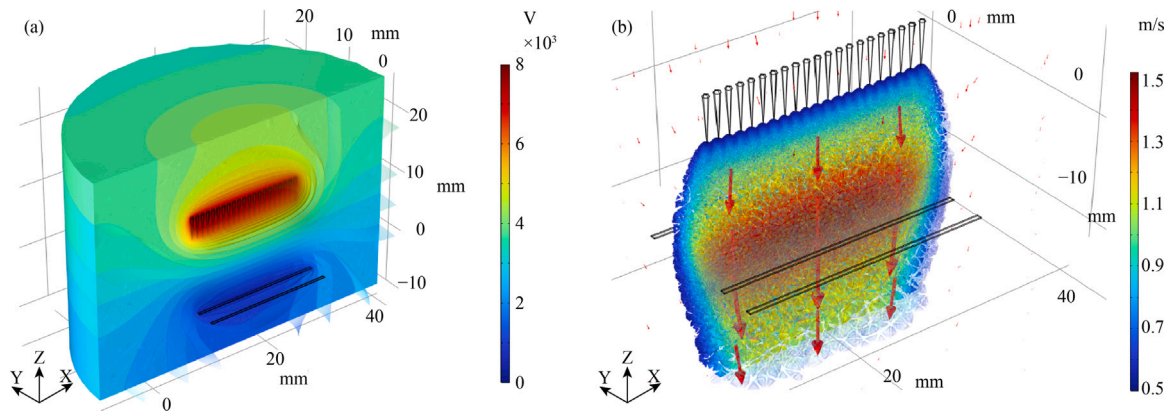


Fig. 8. Distribution of electric potential field and ion wind velocity under the optimized configurations. (a) Electric potential field. (b) Velocity field.

ion wind engines shown in Fig. 9(a), include an LED light source, which was positioned at a distance of 150 cm from the parabolic mirror for the Schlieren photography. Light was reflected from the mirror surface into the camera. The sharp edge of a blade was used to block a portion of the light entering the camera, as illustrated in the enlarged view in Fig. 9(a). Schlieren photography is a physical phenomenon that relies on the deviation of light caused by localized variations in the refractive indices of transparent materials. The light beam emitted from the light source is collimated and passes through the region under measurement,

then is focused by a concave mirror. A knife-edge is placed at the focal point to block the deflected light, thereby creating a contrast between light and dark regions on the imaging plane [49].

At an applied voltage of 7 kV to the emitter, the ion wind airflow generated by the four-collector ion engine was concentrated between the emitter and collectors, as shown in Fig. 9(b), whereas the two-collector ion engine produced no discernible airflow in the Schlieren imaging system. At an increased emitter voltage of 8 kV, the ion wind generated by the ion wind engine with a four-collector configuration

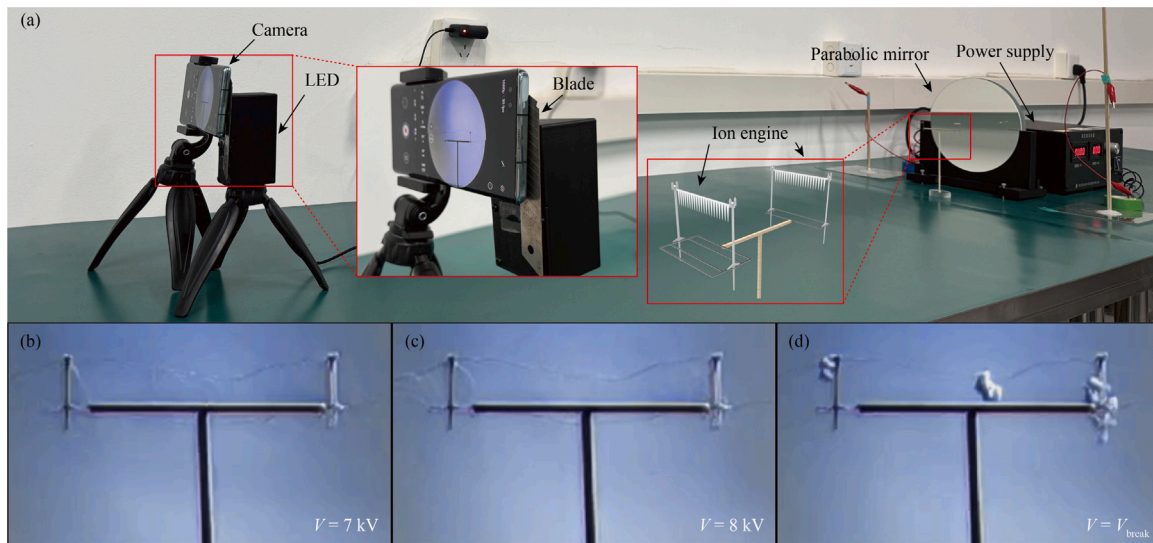


Fig. 9. Test system and results of the schlieren photography. (a) Photograph of experimental tests. Experimental results under (b) 7 kV, (c) 8 kV, and (d) the breakdown voltage.

exhibited a higher velocity and an expanded flow region. At this point, the two-collector ion wind engine began to produce the visible airflow, as shown in Fig. 9(c). But the flow appeared more chaotic compared to that generated by the ion wind engine with four collectors. Upon further voltage increase, electrical breakdown occurred, as illustrated in Fig. 9(d). These results of the Schlieren photography confirm that the structural optimization of the ion wind engine effectively enhances ion wind velocity. Additionally, the four-collector configuration proves advantageous in concentrating the ion wind, which reduces interference when multiple ion wind engines are assembled within a microrobot. This further facilitates the potential for achieving of multi-degree-of-freedom motion control in microrobots equipped with multiple ion wind engines.

Another performance test for the ion wind engine is the measurement of the thrust. The device is shown in Fig. 10(a). First of all, it was necessary to construct an insulating electrostatic shield and place the precision balance inside the shield to prevent the ions generated by the ion wind engine from interfering with the measurement circuit module of the precision balance. At the same time, the ion wind engine and the electrostatic shield needed to be maintained at a certain distance to allow the ion wind to dissipate fully into the air, preventing any impact on the data collection of the precision balance. To achieve this, the engine was fixed on the tip of a long, thin stick. It was important to note that the ion wind engine must be securely fixed to the insulating device to prevent any movement of the thruster while generating ion wind, which could affect the measurements of the precision balance. The thrust of the engine is characterized by the reduction in the reading of the precision balance. The measured thrust is shown in Fig. 10(b). The experimental data were fitted to a curve to characterize the ion wind engine's thrust. As evidenced by the data, when the voltage is below 5 kV, the ion wind engine with four collectors generated thrust, while the ion wind engine with two collectors did not. At the same voltage, the thrust produced by the ion wind engine with four collectors was greater than that of the ion wind engine with two collectors. The optimal ion wind engine generated a thrust of 0.66 mN, with a size of $12 \text{ mm} \times 21 \text{ mm} \times 10 \text{ mm}$, resulting in a thrust per unit volume of 261 N/m^3 , which is higher than that of the Airbus A320 (31.5 N/m^3) and wire-wire ion wind thruster (110 N/m^3) [34]. In addition, an ion wind engine with four collectors weighs 18.1 mg, resulting in a thrust-to-weight ratio of 3.7, which is higher than that of the ion wind engine with two collectors (2.5).

Earlier ion wind engines with larger size were capable of generating thrusts on the order of newtons. For example, Chung et al. [19] developed a wire-to-plane Ionic Flyer that produced 0.16 N of thrust at 28.5 kV (2007), and Xu et al. [1] created a solid-state ionic wind aircraft capable of producing thrusts up to 3.2 N under 40 kV (2018). However, in contrast to large-scale ion wind engines, the thrust of centimeter-scale ion wind engines is typically on the millinewton (mN) scale, and their operating voltages are generally lower. Recent studies on centimeter-scale small ion wind engines have focused on miniaturization, increasing thrust, and reducing mass. For instance, Mai et al. [3] developed an ion-wind-powered boat that achieved a thrust of 4 mN at 17 kV (2025), while Liu et al. [6] developed an ion wind engine with needle-to-ring configuration that produced 7.04 mN of thrust at 20 kV (2025). These studies highlight the thrust potential of ion wind engines. However, the engines developed in these works have larger sizes, greater mass, and higher operating voltages compared to the structure proposed in this paper. The structure we propose is more compact, with a thrust per volume comparable to that of the needle-to-ring configuration, but it operates at a lower voltage. Drew et al. [20] proposed a silicon-based, four-engine ion wind microrobot, achieving a total thrust of 1 mN (with an average thrust of 0.25 mN per engine, 2018). Zhang et al. [36] designed an ion wind engine producing a thrust of 0.7 mN (2021), which is similar to our proposed structure, but their collector area ($19 \text{ mm} \times 30 \text{ mm}$) is 2.2 times larger, and its mass is 2.3 times greater than ours, resulting in a lower thrust-to-area ratio compared to our design. In 2024, Drew et al. [35] introduced a multi-stage ion wind engine that increased the single-stage thrust of approximately 0.5 mN to 3 mN. While this is a promising approach for higher thrust, it should be noted that increasing the structure's height could negatively impact flight stability. In comparison, our proposed design uses PET film with an aluminum coating that has been processed using only an ultraviolet laser cutter. Although a classic needle-to-plane configuration is used, the electrode distribution design staggers the emitter and collector, causing the generated ion wind to primarily flow in the middle of the collectors. This design reduces drag force, thereby enhancing thrust [2]. Additionally, the auxiliary collectors on the outer sides help extend the discharge [30]. As a result, by using inexpensive materials and processing methods, along with a compact structure, sufficient thrust can be generated to propel the microrobot into flight. This method is more cost-effective than photolithography-based structures [20]. Its compact structure and lightweight design also make it a promising candidate for microrobotic applications.

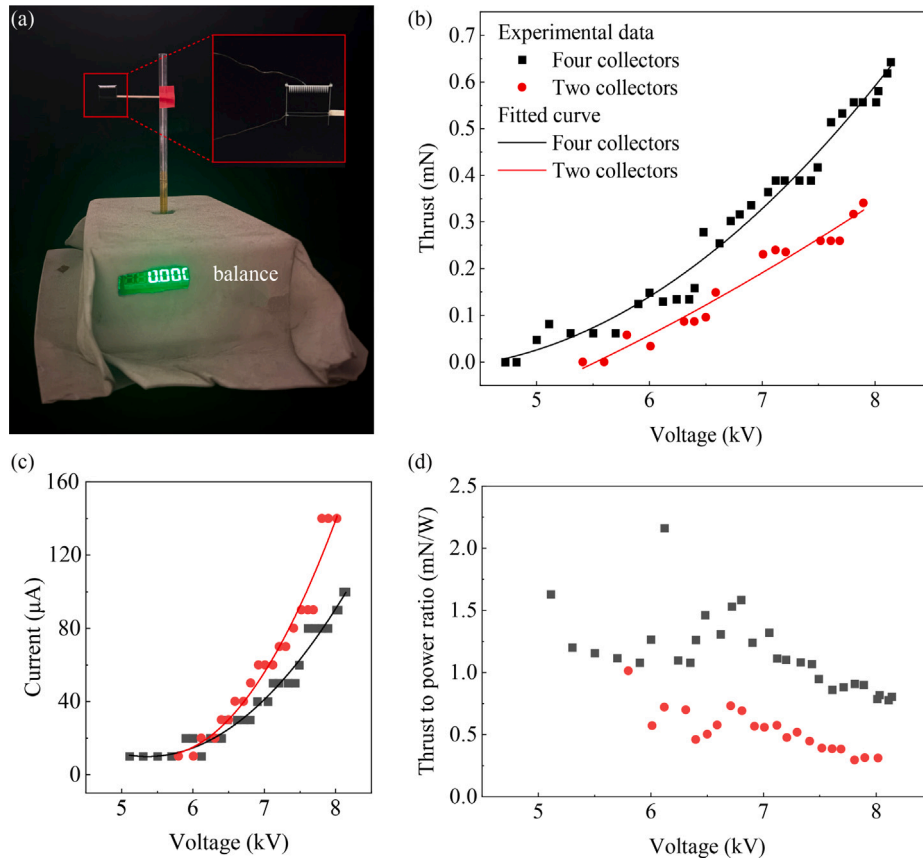


Fig. 10. Variation of thrust, current, and thrust-to-power ratio with applied voltage for ion wind engines under different collector configurations. (a) The experimental setup. (b) Thrust versus applied voltage. (c) Current versus applied voltage. (d) Thrust-to-power ratio versus applied voltage.

During the thrust measurement process, the current was also recorded, as shown in Fig. 10(c). Then the thrust-to-power ratio is calculated, as shown in Fig. 10(d). The thrust-to-power ratio of the engine with four collectors is slightly higher than that of the engine with two collectors, reaching up to 2.1 mN/W in low-thrust conditions. As the thrust increases, the thrust-to-power ratio decreases, with a value of approximately 0.8 mN/W at high thrust. This thrust-to-power ratio is slightly lower than that reported by Drew et al. (1.86 mN/W, [35]) and comparable to that reported by Liu et al. (1.0 mN/W, [6]). In the experiments, energy was supplied via a 50-micrometer copper wire, with the copper wire connected to the electrode using conductive silver glue. The energy loss in this part could be further optimized. Further research is needed to improve energy efficiency. This may include material substitutions, the design of more efficient power supply circuits, and other optimizations.

4. Application showcase: flying microrobot

4.1. Design and simulation

The integration of multiple ion wind engines into a microrobotic system can introduce multi-degree-of-freedom motion capabilities. The optimized engine arrangement can also enhance flight stability. In order to achieve a more optimized multi-engine microrobot, two distinct configurations were designed: an equilateral triangle structure with a 60° angle between adjacent sides and a quadrilateral structure with a 90° angle between adjacent sides. The triangular configuration, with fewer propulsion units, forms a stable triangular support between the sides. In contrast, the quadrilateral structure offers greater flexibility, allowing for multi-directional, high-mobility, and decoupled movement through the adjustment of the thrust generated by the individual engine.

Simulations were conducted for both configurations of the microrobot, and the three-dimensional electrical potential and velocity distributions of the triangle and quadrilateral ion wind flying microrobots are shown in Fig. 11. The electrical potential and velocity distributions of one engine on one side are similar to those observed in the simulation of a single engine. However, there are differences in the overall wind velocity and electrical potential of the microrobot with the different configurations. In the quadrilateral configuration, the high voltage is distributed around the edges, creating a larger central low-pressure region. This results in a broader area with faster ion wind flow. As can be seen in the velocity distribution chart on the right side of Fig. 11, the maximum ion wind speed of the quadrilateral microrobot reaches 1.67 m/s, compared to 1.56 m/s for the triangle structure, showing an improvement. Although the ion wind generated by the ion engine is primarily concentrated in the central portion of the collector structure, there is also a notable macroscopic flow with a certain velocity present on the outer side of the collector. This region is defined as the secondary flow region. As can be seen in Fig. 11(b) and (d), the secondary flow regions generated by multiple ion engines overlap at the structural center of the triangle and quadrilateral ion wind flying microrobots. The maximum ion wind speed in the secondary flow region formed by the quadrilateral structure is 1.208 m/s, which is an 80% increase over the secondary flow speed in the triangular configuration. This significant improvement in the ion wind speed in the secondary flow region has a marked impact on the overall ion wind velocity, greatly enhancing the thrust and stability of the ion wind flying microrobot.

The feasibility of generating unbalanced thrust by independently controlling the engines of a quadrilateral microrobot was further investigated through simulation analysis. In the design of ion wind engines for microrobots, it is crucial to account for potential interference between multiple ion engines when electromagnetic shielding and flow

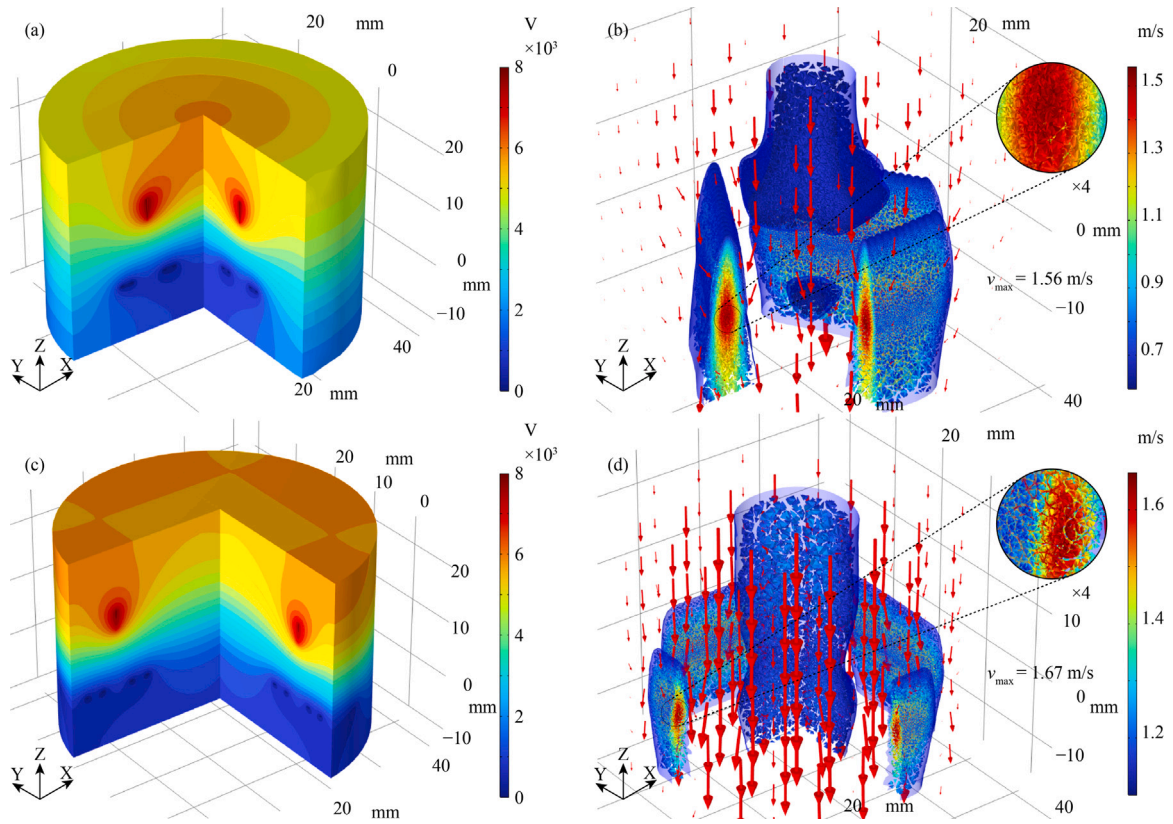


Fig. 11. Electric potential field and velocity distribution of microrobot with three and four ion wind engines. (a) Electric potential field and (b) velocity field of the microrobot with three ion wind engines. (c) Electric potential field and (d) velocity field of the microrobot with four ion wind engines.

isolation are not incorporated. Without these design considerations, the electric fields and airflow generated by neighboring engines may interact, potentially altering the intended ion wind's strength and direction for each engine. This interaction could significantly affect the overall propulsion performance. To ensure that each ion engine operates independently, it is necessary to verify that the ion wind generated by each engine is primarily controlled by the voltage applied to the emitter and collector of the engine, with minimal or no influence from adjacent engines. Such validation is essential for achieving precise and controlled propulsion in multi-engine configurations. In the calculations, the emitter on the side corresponding to the negative X -axis direction was grounded. The results are shown in Fig. 12. By comparing these results with the voltage distribution of the fully powered quadrilateral microrobot (Fig. 11(b)), it is observed that within the same applied voltage, the high-voltage regions are decreasing in size. This caused the high-potential distribution of the quadrilateral microrobot to shift toward the emitters of the remaining three engines. Furthermore, due to the grounding of the emitter of one engine, the central ion wind flow zone is getting smaller, and the ion wind speed near the grounded side decreases. However, the ion wind speed on the opposite side remained unaffected. In the simulation, the ion wind velocity between the electrodes of the unpowered side does not exceed 0.3 m/s, which can be considered negligible in terms of contributing thrust to the microrobot. The calculation results indicate that grounding the emitter on one side reduces the thrust on that side, but does not affect the thrust generation of the other engines. Consequently, this configuration enables the microrobot to achieve multi-degree-of-freedom movement by independently controlling the thrust produced by each engine without a special electromagnetic shielding design.

4.2. Experimental validation

To verify the flight capability of the multi-ion-engine microrobots proposed in the simulation, a quadrilateral flying microrobot was developed, as shown in Fig. 13(a). Firstly, the emitters of all engines were connected to a positive high voltage (HV), and the collectors were all grounded. The settings were consistent with the simulation setup. A comparison of Fig. 13(b) and (c) reveals that when the voltage reached 7.7 kV, the ion wind engine's EHD thrust effectively counterbalanced the microrobot's weight, enabling successful takeoff.

Furthermore, experiments were conducted to validate the engine's capability for independent thrust control. As in the simulation setup, the emitter of one engine was grounded. Due to the absence of thrust on one side, a voltage of 7.7 kV was not sufficient to lift the microrobot. The emitter voltage was then increased on the remaining three engines. When the voltage was raised to 8.0 kV, the microrobot took off but tilted toward the left, as shown in Fig. 13(d). When the thrust on one side (F_1) is smaller than that on the opposite side (F_2), the force generates a moment, causing the microrobot to rotate around axis X_b , and leading to a change in its orientation angle. The flight experiments (Movie S1) demonstrate that the designed microrobot is capable of adjusting its attitude by precisely controlling electrical energy inputs.

Additionally, we conducted further experiments to control the flight attitude by altering the voltage differential applied to the engine, thereby adjusting the engine's thrust. A negative high-voltage power supply and a high-voltage amplifier were used. The specific circuit is shown in Fig. 14(a). The negative high-voltage power supply was connected to the collector, applying -9 kV to it. The high-voltage amplifier applied a voltage between 0 and -1 kV to the emitters, with both power supplies sharing a common ground. For example, when the adjustable high-voltage amplifier applied 0 V, the voltage differential between the emitters and the collector was 9 kV. When the

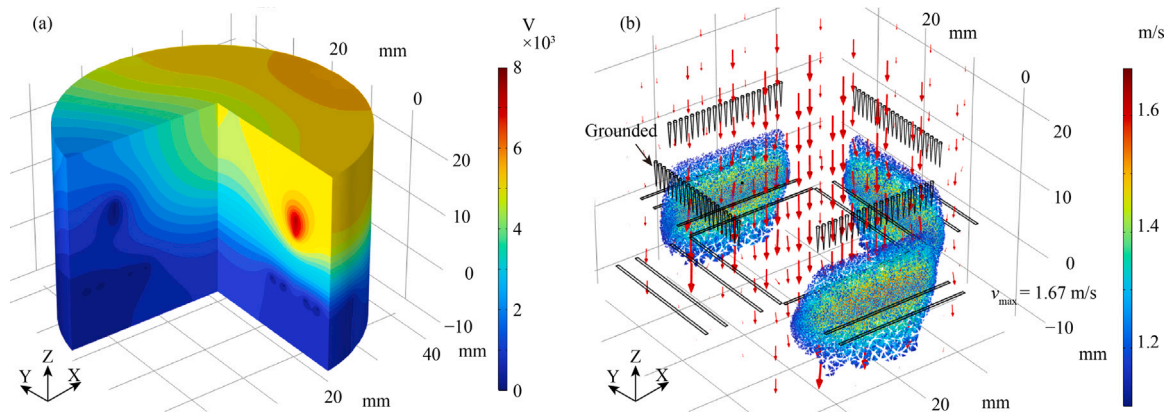


Fig. 12. Electric potential field and velocity distribution when the voltage of the ion wind engine is controlled independently. (a) Electric potential field. (b) Velocity field.

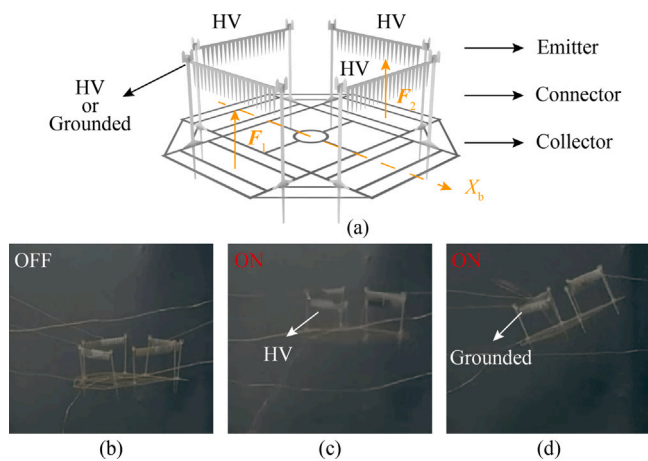


Fig. 13. Experimental results of the microrobots. (a) Illustration of the micro-robot. (b) HV is off. (c) HV is all on. (d) One of the emitters is grounded.

adjustable high-voltage amplifier applied -1 kV, the voltage differential was reduced to 8 kV, resulting in less thrust compared to the 9 kV differential. The torque ($M = \Delta F l$) generated by the difference in thrust causes the microrobot to tilt toward the engine with the 8 kV voltage differential, as shown in Fig. 14(b). Furthermore, we conducted an experiment to simultaneously control two adjacent engines, as shown in Fig. 14(c). The results of this experiment are presented in Fig. 14(d). The experiments (Movie S2) further confirm the feasibility of controlling the microrobot's attitude by adjusting the voltage differential and the resulting thrust.

We also tested the load-carrying capacity of the microrobot. Based on the optimization of the ion wind engine, the ion-propelled microrobot developed in this study, weighing 89 mg, successfully completed its flight while carrying a payload of 257 mg (Fig. 15, Movie S3), achieving a thrust-to-weight ratio of 3.8 .

5. Conclusion

We systematically enhanced the thrust characteristics of ion wind engines through a tightly coupled numerical and experimental methodology. These optimized thrusters enabled the propulsion of both triangular and quadrilateral microrobot platforms. Crucially, independent actuation of a single thruster provided precise attitude control, a critical step for achieving controllable flight of a solid-state aerial microrobot. The principal findings of this work are as follows:

- (1) Increasing the spacing between the emitters and collectors, applying higher voltages to the emitters, reducing the emitter tip angle, increasing the number of needle tips of the emitter, and optimizing the arrangement of collectors have all been shown to effectively enhance ion wind velocity. These factors, when strategically combined, can significantly improve the performance of the ion wind engine, thereby contributing to higher thrust.
- (2) The optimized ion wind engine has a compact size of 2520 mm³, yet it can generate a thrust of 66 mN. With a thrust per unit volume of 261 N/m³, it is well-suited for use as a propulsion unit in microrobots. Moreover, the ion wind engine with four collectors weighs only 18.1 mg, yielding a thrust-to-weight ratio of 3.7 , which is higher than that of the ion wind engine with two collectors by 48% .
- (3) It was observed that polygonal microrobots equipped with the optimized ion wind engines generate higher ion wind velocities as the number of sides increases, due to the formation of larger and faster flow regions near the center of the robot. Appropriate structural design enables multiple ion wind engines to operate independently, thereby achieving the desired motion.

CRediT authorship contribution statement

Qiannan Tao: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Zhi Chen:** Writing – original draft, Validation, Methodology, Investigation. **Yunqi Cao:** Writing – review & editing, Methodology. **Wei Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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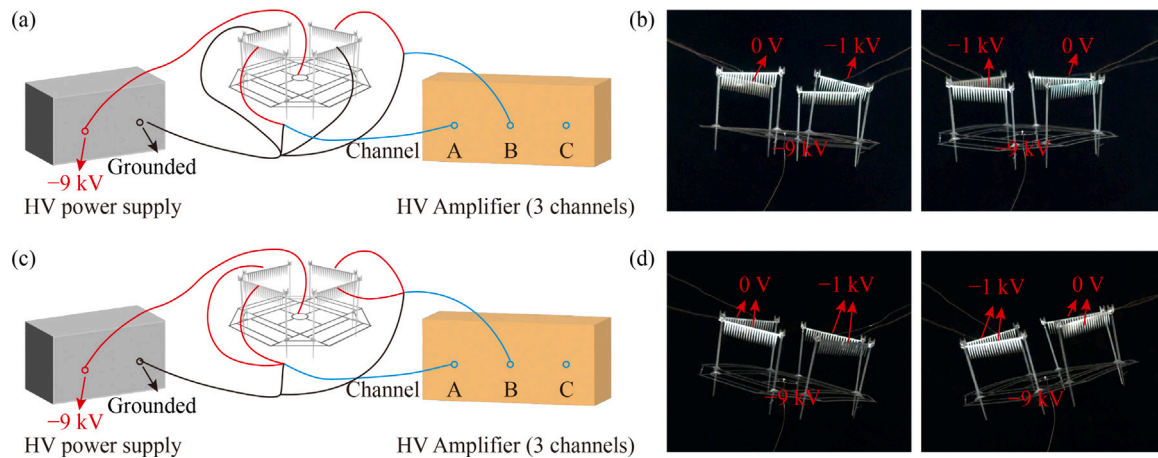


Fig. 14. Experimental results of the microrobots controlled by an HV amplifier. (a) Circuit diagram for controlling two engines simultaneously. (b) Attitude adjustment of the microrobot with two engines under simultaneous control. (c) Circuit diagram for controlling four engines simultaneously. (d) Attitude adjustment of the microrobot with four engines under simultaneous control.

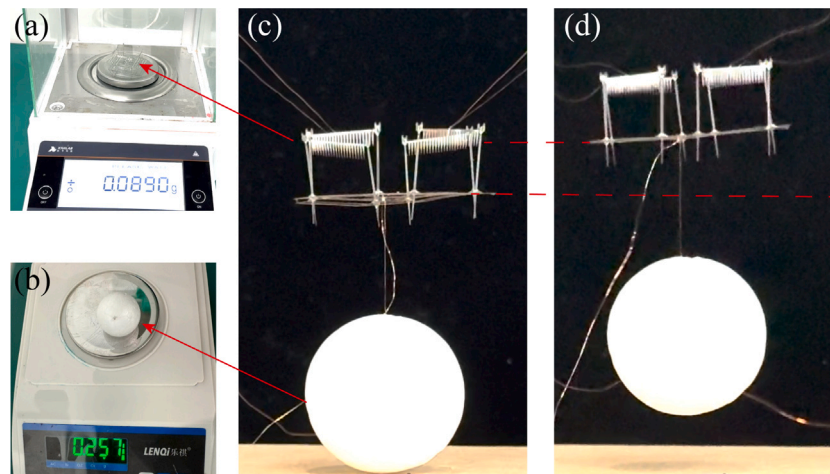


Fig. 15. Load test of the microrobot. (a) The microrobot weighs 89 mg. (b) The payload weighs 257 mg. (c) The microrobot is equipped with the payload. (d) The microrobot hovers in the air with a load of 2.8 times its own weight.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.energy.2025.139027>.

Data availability

Data will be made available on request.

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